How-To Library

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Using Conditional Constraints in Data Generation

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Abstract

In Test Data Manager, you can create and assign data generation rules to generate test data in the target columns. Create conditional constraints to restrict the test data that you want to generate in a target column based on the data in another column. This article describes how to create and use a random date generation rule and a conditional constraint in Test Data Manager to generate test data.

Supported Versions

- Test Data Management 9.6.0 - 9.6.1

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Overview

In Test Data Manager, create and assign a random date generation rule to a target column. Create and use a conditional constraint to restrict the data that you want to generate in a column based on the data in another column. Configure an expression to apply constraints at the table level. Create a data generation plan, and run the workflow to generate realistic test data in the target.

Scenario

You work for an organization in the software testing department, and you want to test an application that maintains joining and confirmation dates of employees. To test this application, you need to generate sample data in the joining date and confirmation date columns of the target table. Create and assign a data generation rule to generate dates in the target columns. You can provide a conditional constraint that confirmation date must be later than the joining date.

To generate the test data, perform the following tasks:

1. Create a random date generation rule.
2. Create a project and import target metadata.
3. Add the rule to the project.
4. Assign the rule to the target columns.
5. Create a conditional constraint.
6. Create a data generation plan.
7. Manage the workflow.

Step 1. Create a Random Date Generation Rule

Create a random date generation rule to generate dates in the target columns.

1. In Test Data Manager, click **Policies**.
2. Click **Actions** > **New** > **Generation Rule**.

   The **Rule Wizard** appears.
3. Enter a name for the rule.
4. Optionally, enter a description for the rule.
5. Select the **Date** data type.
6. Choose **Standard Rule** and select **Random**.
7. To override the data generation parameters for a rule, select **Override Allowed**.
8. Click **Next**.
9. Configure the following data generation parameters:

   Start Date/Time

   Enter the start date and time for the generated dates.

   End Date/Time

   Enter the end date and time for the generated dates.

   Exclude

   Optional. Enter the days of the week that you do not want to include in the generated dates. You can enter up to two days.

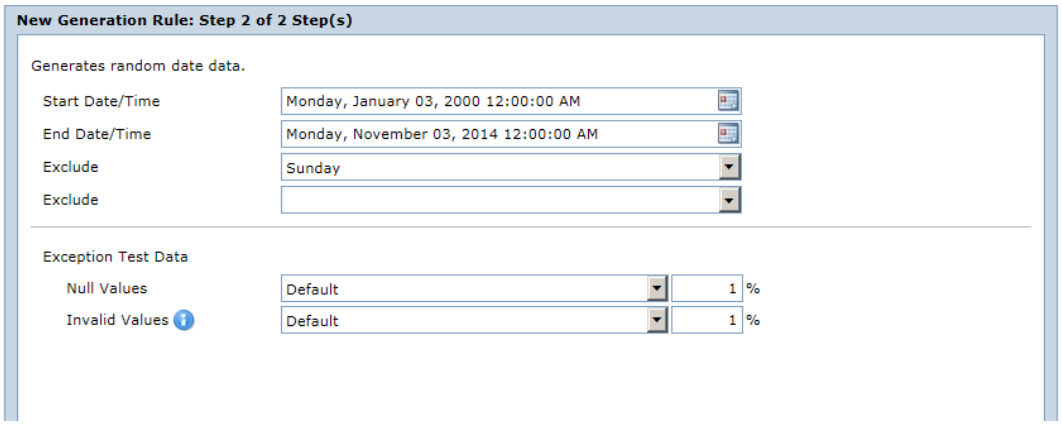
   Null Values

   Optional. To generate null values, enter a percent value.

   Invalid Values

   Optional. To generate values that are not valid, enter a percent value.

   The following image shows the random date generation rule parameters:



10. Click **Finish**.

    The rule appears in the list of generation rules.

Step 2. Create a Project and Import Target Metadata

In Test Data Manager, create a project folder to store the data generation components such as entities and tables. Import target metadata into the project. In this example, the target database is AutoTarget, the target table is EMPLOYEE_RECORD, and the target columns are DATE_OF_JOINING and DATE_OF_CONFIRMATION.

1. Click **Projects**.

   A list of projects appears.
2. Click **Actions** > **New**.
3. In the **Create Project** dialog box, enter the following project properties:

   Name

   Enter a name.

   Description

   Enter a description.

   PowerCenter Repository

   The default is the name of the PowerCenter repository.

   Folder

   The default is the project name.

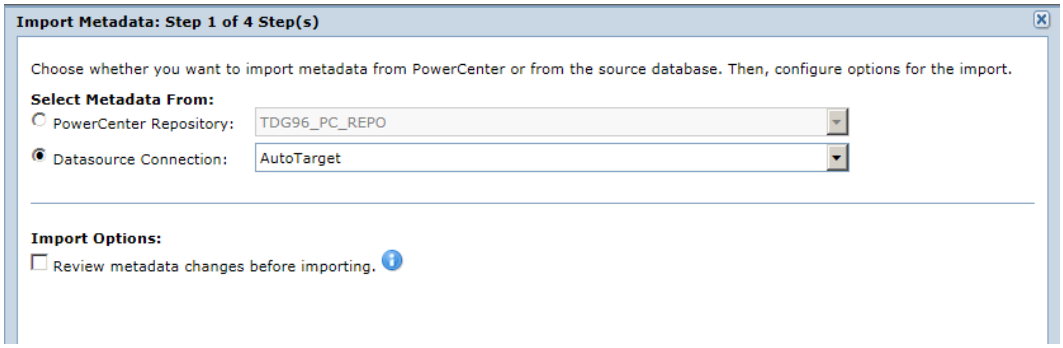
   Owner

   The default is the name of the user that created the folder.
4. Click **OK**.

   View the properties of the project that you created.
5. Click **Actions** > **Import Metadata**.

   The **Import Metadata** dialog box appears.
6. To import metadata from a database, choose **Datasource Connection** and select the database.

   The following image shows the options to import metadata:



7. Choose whether you want to review the metadata changes before you import the data sources. You can choose to skip the import options.
8. Click **Next**.
9. Select the metadata schema that you want to import from the database. You can filter folders by folder name or description.
10. Click **Next**.
11. Select the tables that you want to import.
12. Click **Next**.

13. To import the data source immediately, select **Import Now**.
14. Click **Finish**.

    While the import runs, view the progress of the import job in the **Monitor** view. After the import job finishes, you can access the imported metadata through the **Data Sources** details view.

Step 3. Add the Rule to the project

You must add the random date generation rule to the project so that you can assign the generation rule to the target columns.

1. Click **Projects**.

   A list of projects appears.
2. Open the project that you created.

   The project window opens in another tab.
3. In the project, click **Overview** > **Policies**.
4. Click **Actions** > **Add Additional Rules**.

   The **Add Additional Rules** dialog box appears.
5. From the list, select the random date generation rule that you created.
6. Click **OK**.

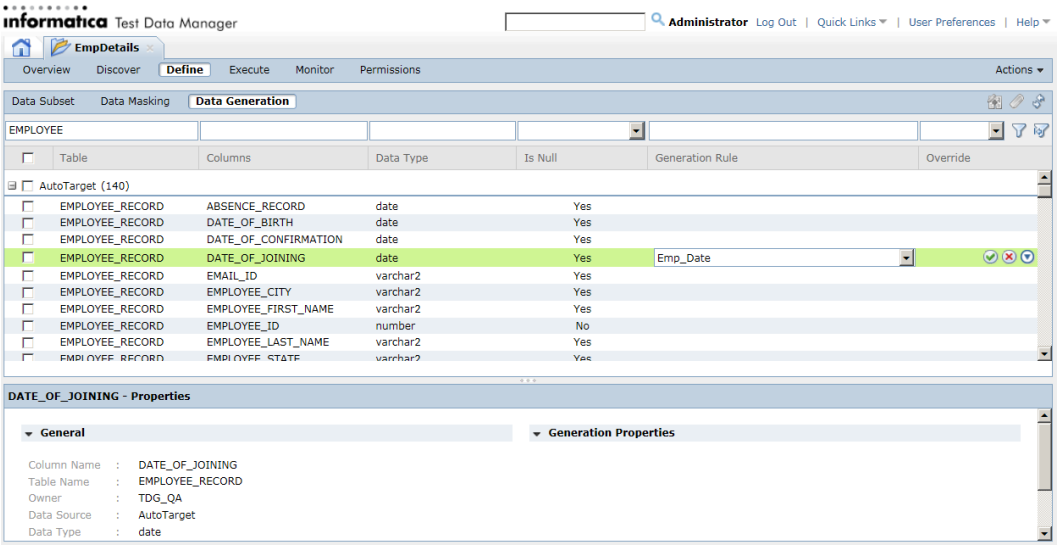
   The rule that you created appears under the **Additional Rules** list.

Step 4. Assign the Rule to the Target Columns

Assign the random date generation rule to the target columns in which you want to generate the test data.

1. In the project, click **Define** > **Data Generation**.
2. In the target table, select the target column where you want to generate the joining dates.
3. Click inside the **Generation Rule** column.
4. Select the random date generation rule that you created.

   The following image shows a sample generation rule assignment for the date data type:



5. To save the rule assignment, click **Save**.
6. Select the target column where you want to generate the confirmation dates.
7. Click inside the **Generation Rule** column.
8. Select the random date generation rule that you created.
9. To save the rule assignment, click **Save**.

Step 5. Create a Conditional Constraint

To create a conditional constraint, configure an expression to restrict the data generated in a column based on the data in another column.

1. In a project, click **Discover** > **Tables**.
2. Click the table in which you want to create the conditional constraint.
3. Click the **Constraints** tab.
4. Click **Create New Constraint**.

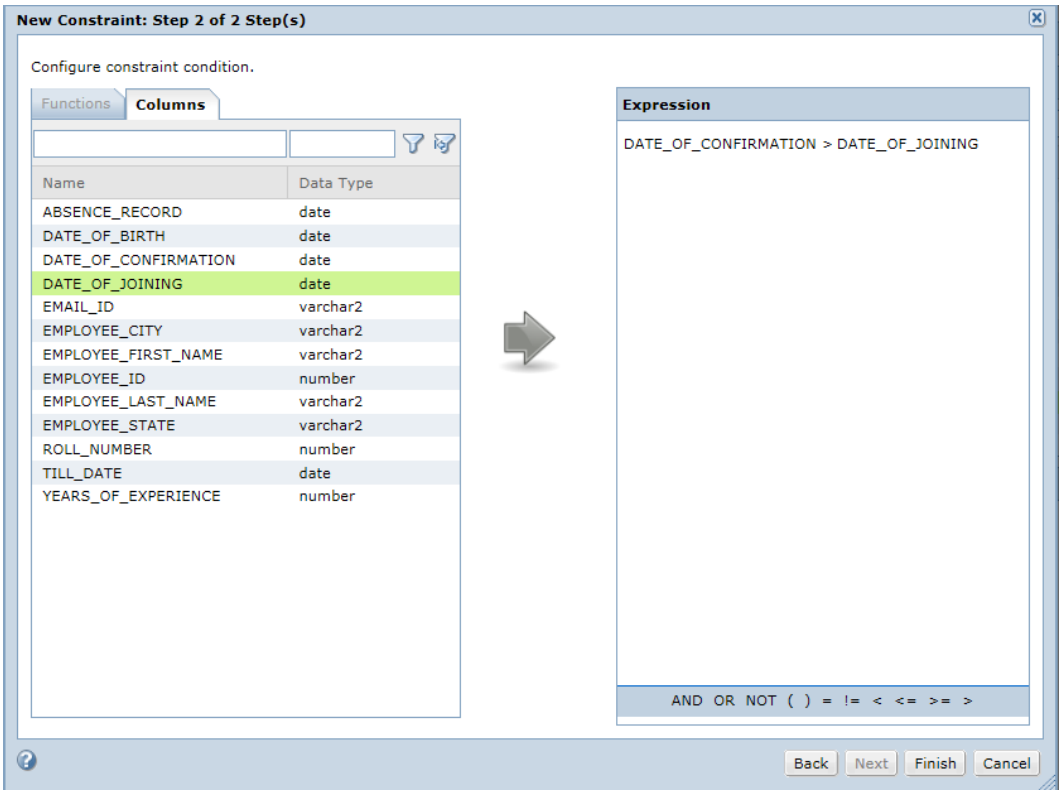
   The **New Constraint** dialog box appears.
5. Enter a name for the conditional constraint.
6. Select Conditional constraint type.

   When you select conditional constraint type, the Severity Level and Parent Table fields become disabled.
7. Select **Enable Constraint**.
8. Click **Next**.
9. Create an expression in the **Expression Builder**. Select the required columns, functions, variables, and operators to build expressions.

   As per the example, you can enter the following conditional constraint:

   ```
   DATE_OF_CONFIRMATION > DATE_OF_JOINING
   ```

The following image shows the **Expression Builder** within the **New Constraint** dialog box:



10. Click **Finish**.

Step 6. Create a Data Generation Plan

To perform the data generation operation, you must create a plan. The plan includes data generation components that you need to generate a workflow.

1. In the project, click **Execute**.
2. Click **New Plan**.
3. In the **New Plan** dialog box, enter a name for the plan.
4. Optionally, enter a description for the plan.
5. Click **Next**.
6. To skip adding the masking components, click **Next**.
7. To skip adding the subset components, click **Next**.
8. To add the data generation components, click **Add Generation Components**.
9. Select the target table to add to the plan.
10. Click **OK**.

    The target table appears in the list.
11. Click **Next**.
12. Review the data generation component and table criteria.
13. Click **Next**.
14. To generate the data, configure the following plan settings:

Number of Records to Generate

Enter the number of records that you want to generate.

Target Connection

Select **Relational**. Click **Select** and choose the target database.

Truncate Tables

To truncate the table before the data loads, select **Truncate Tables**.

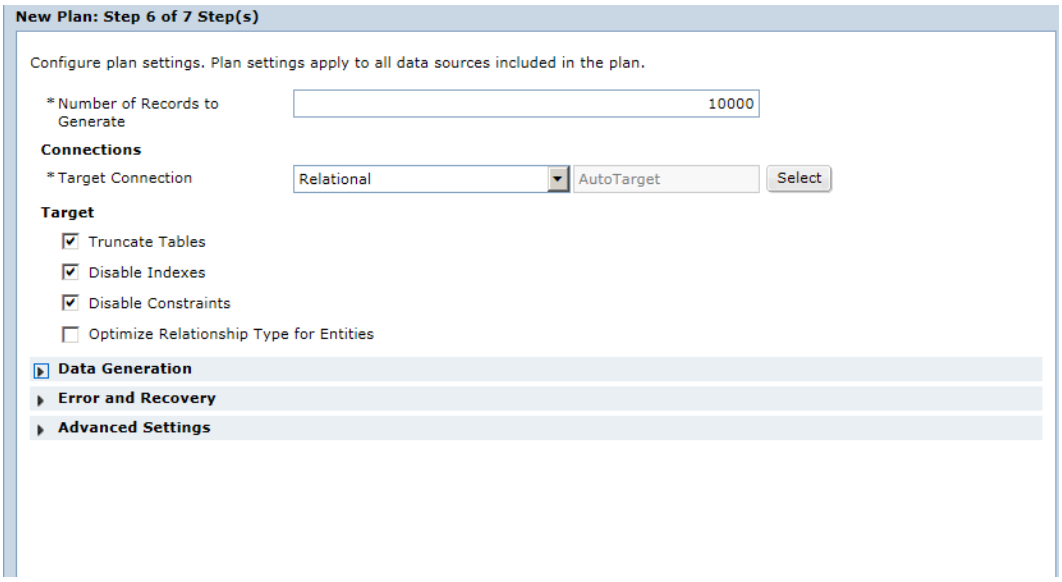
Disable Indexes

To disable indexes for faster loading of data, select **Disable Indexes**.

Disable Constraints

To disable physical constraints before the data is loaded to the target database, select **Disable Constraints**. After the data loads, the constraints are enabled automatically.

The following image shows sample settings for a data generation plan:



15. Click **Next**.
16. Review the data source settings.
17. Click **Finish**.

Step 7. Manage the Workflow

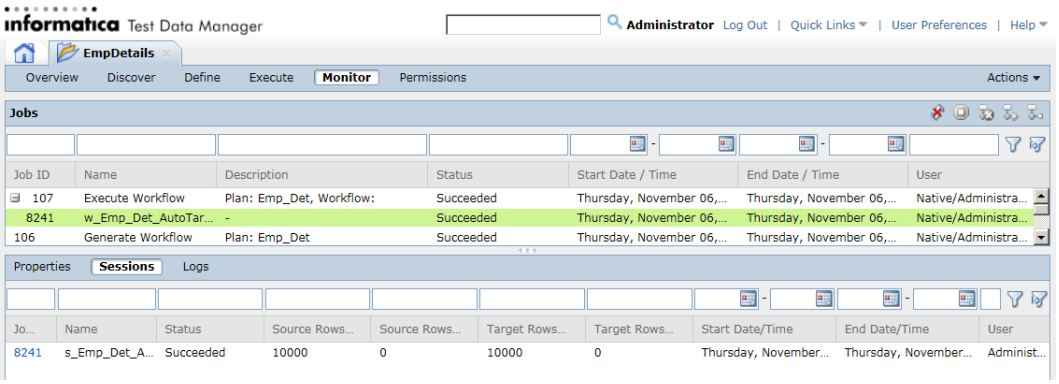
After you create the plan, you must run the workflow to generate the test data in the target table. You can monitor the progress of the workflow jobs. When the workflow job succeeds, TDM generates data in the target table.

1. To access the plan in the project, click **Execute**.
2. Select the data generation plan that you created.
3. Click **Actions** > **Generate and Execute**.

   The **Generate and Execute Plan** dialog box appears.
4. Select the PowerCenter Integration Service.
5. Select **Schedule Now**.

6. Click **Execute Workflow**.
7. To view the status of the workflow jobs, click **Monitor**.
8. To refresh the view, click **Auto Refresh On**.
9. To view the job details, select the job ID and click the **Properties**, **Sessions**, and **Logs** tabs.

   The following image shows the status of the sample workflow jobs:



Author

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